

Tyler Roberts

Senior Software Engineer

CONTACT

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SKILLS

Programming

Python, TypeScript, JavaScript, Node.js

AI_ML

Large Language Models (LLMs), Prompt Engineering, Embeddings, Semantic Search, Retrieval-Augmented Generation (RAG), Agentic AI Systems, LLM Evaluation Pipelines

Vector_Databases

Weaviate, Chroma, pgvector

Frameworks_Tools

LangChain, LlamaIndex, Pydantic AI, DSPy

Cloud_DevOps

AWS, ECS, Lambda, S3, Docker, CI/CD, Observability

Frontend_Integration

React, IDE Extensions, VS Code, MCP Integrations

EDUCATION

Florida State University

Bachelor of Computer Science
2009 - 2013

Professional Summary

Senior Software Engineer with 9+ years of experience designing and shipping production-grade AI-powered and cloud-native systems. Strong background in building LLM-based applications, semantic search, embeddings pipelines, and agentic AI workflows integrated into real-world products. Proven ability to own AI features end-to-end, from data ingestion and retrieval design to scalable inference and deployment on AWS. Experienced in Python-first ML development with supporting TypeScript services and modern DevOps practices. Adept at collaborating with domain experts to ensure AI outputs are accurate, reliable, and grounded in technical correctness.

Professional Skills

Senior Software Engineer

Web Venturez LLC | February 2020 - October 2025

- Designed and delivered production-grade AI-powered features using LLMs, embeddings, and semantic search to enhance end-user workflows.
- Built and maintained retrieval pipelines over large technical datasets using vector databases and optimized chunking strategies.
- Implemented agentic AI systems coordinating multiple tools and models to deliver accurate, context-aware responses.
- Owned end-to-end ML services from experimentation through production inference and monitoring.
- Developed scalable backend services in Python and Node.js to support AI-driven functionality.
- Integrated LLM capabilities into user-facing applications and developer tooling with a strong focus on usability.
- Optimized inference latency, quality, and cost through prompt tuning, caching, and model selection strategies.
- Deployed containerized services on AWS using ECS, Lambda, and S3 with infrastructure-as-code practices.
- Introduced structured logging, metrics, and observability to improve reliability of AI services.
- Collaborated closely with product managers and domain experts to ensure AI outputs were technically correct.
- Served as a technical owner for complex integrations and critical AI components across the platform.

Full Stack Engineer

Webiz Digital | July 2017 - January 2020

- Developed full-stack applications combining Python and Node.js backends with modern React frontends.

- Designed REST APIs and backend services supporting data-heavy and logic-intensive workflows.
- Contributed to early AI and search-related features by integrating external APIs and data pipelines.
- Worked with cloud-hosted environments to deploy and maintain scalable services on AWS.
- Collaborated with designers and stakeholders to translate requirements into reliable implementations.
- Improved application performance and maintainability through refactoring and code reviews.
- Supported production systems by debugging issues and improving error handling.
- Assisted with containerization and deployment workflows using Docker.
- Built a strong foundation in distributed systems and service-oriented architecture.

Software Developer

Centric Tech Inc. | October 2013 - June 2017

- Developed backend services and web applications using JavaScript and related technologies.
 - Implemented core business logic and APIs for internal and client-facing systems.
 - Worked closely with senior engineers to understand system architecture and deployment pipelines.
 - Participated in debugging, performance tuning, and production support activities.
 - Contributed to early distributed system designs and service integrations.
 - Collaborated with cross-functional teams to deliver features on schedule.
 - Established strong software engineering fundamentals and best practices.
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